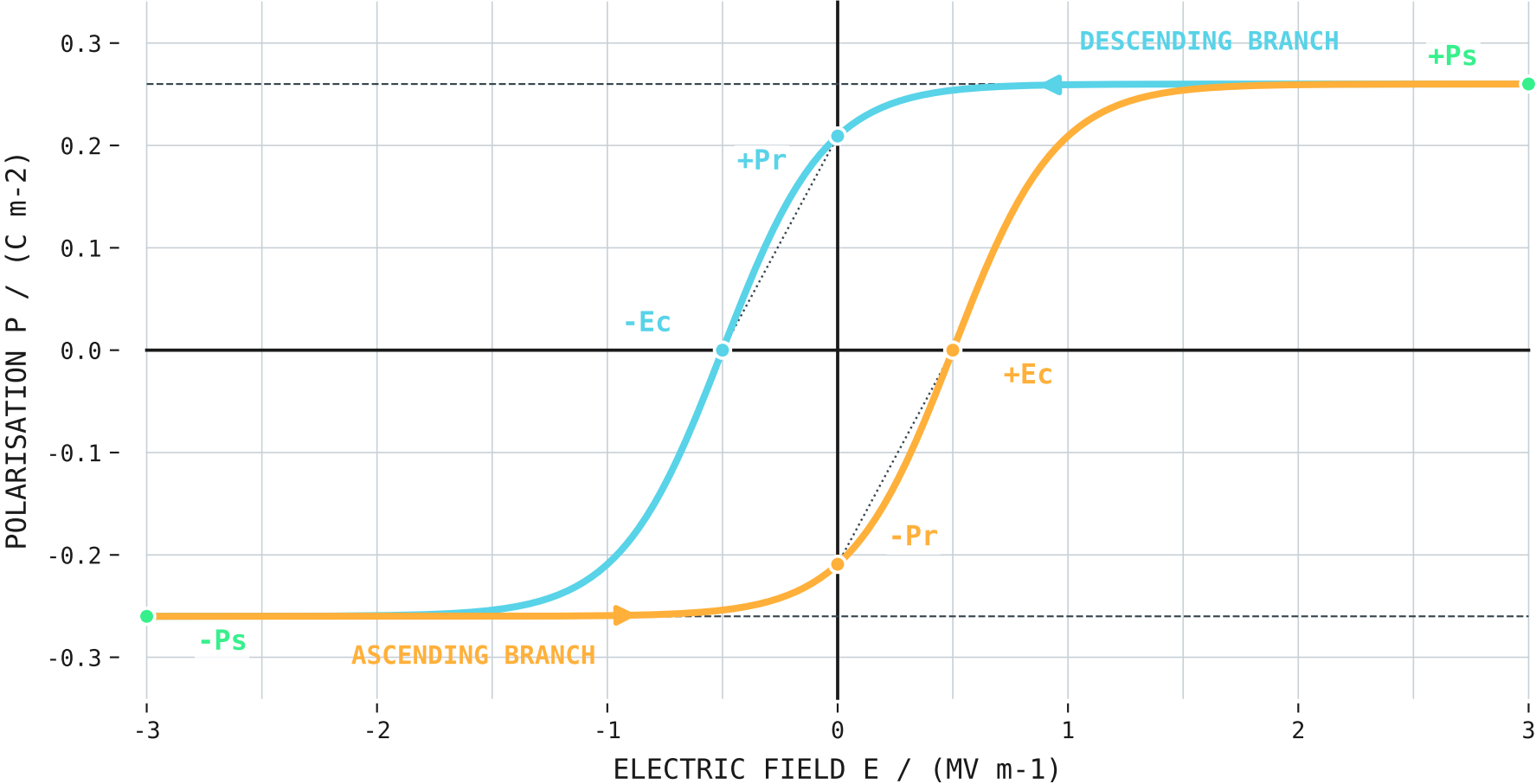


BaTiO3 FERROELECTRIC HYSTERESIS LOOP : TWO BRANCHES, P = 0 AT PLUS AND MINUS Ec



[MODEL] TANH SWITCHING MODEL DRAWN TO SCALE FROM $P_s = 0.26 \text{ C m}^{-2}$, $P_r = 0.21 \text{ C m}^{-2}$ AND $E_c = 0.5 \text{ MV m}^{-1}$; NOT MEASURED DATA

[Ti] THE RIGID-ION Ti-ONLY SUM OF THIS ASSIGNMENT GIVES ONLY 0.0805 C m^{-2} , SO THE MEASURED P_s IS USED FOR THE LOOP HEIGHT

[Ec] REAL CERAMIC E_c IS $1e5$ TO $1e6 \text{ V m}^{-1}$ BECAUSE 180-DEGREE DOMAIN WALLS SWEEP; HOMOGENEOUS REVERSAL WOULD NEED ABOUT $1e8 \text{ V m}^{-1}$